



# Prayas JEE 2.0 (2024)

## Quadratic Equations

**DPP-01**

- If the equation  $ax^2 - 5x + c = 0$  has 10 as the sum of the roots and also as the product of the roots, then
  - $a = \frac{1}{2}, c = 5$
  - $a = 2, c = 3$
  - $a = 5, c = \frac{1}{2}$
  - $a = 3, c = 2$
- Let  $\alpha$  and  $\beta$  are the roots of the equation  $x^2 + x + 1 = 0$  then
  - $\alpha^2 + \beta^2 = 4$
  - $(\alpha - \beta)^2 = 3$
  - $\alpha^3 + \beta^3 = 2$
  - $\alpha^4 + \beta^4 = 1$
- If  $\alpha$  and  $\beta$  are the roots of the equation  $x^2 + 6x + \lambda = 0$  and  $3\alpha + 2\beta = -20$ , then  $\lambda =$ 
  - 8
  - 16
  - 16
  - 8
- If the roots of  $x^2 - bx + c = 0$  are two consecutive integers, then  $b^2 - 4c$  is
  - 0
  - 1
  - 2
  - None of these
- The value of 'a' for which one root of the quadratic equation  $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$  is twice as large as the other is
  - $-\frac{1}{3}$
  - $\frac{2}{3}$
  - $-\frac{2}{3}$
  - None of these
- Difference between the corresponding roots of  $x^2 + ax + b = 0$  and  $x^2 + bx + a = 0$  is same and  $a \neq b$ , then
  - $a + b + 4 = 0$
  - $a + b - 4 = 0$
  - $a - b - 4 = 0$
  - $a - b + 4 = 0$
- Sachin and Rahul attempted to solve a quadratic equation. Sachin made a mistake in writing down the constant term and ended up in roots (4, 3). Rahul made a mistake in writing down coefficient of  $x$  to get roots (3, 2). The correct roots of equation are:
  - 4, -3
  - 6, 1
  - 4, 3
  - 6, -1
- If the sum of the roots of the equation  $ax^2 + bx + c = 0$  be equal to the sum of their squares, then
  - $a(a + b) = 2bc$
  - $c(a + c) = 2ab$
  - $b(a + b) = 2ac$
  - $b(a + b) = ac$
- If  $\alpha, \beta$  are the roots of quadratic equation  $x^2 + px + q = 0$  and  $\gamma, \delta$  are the roots of  $x^2 + px - r = 0$ , then  $(\alpha - \gamma) \cdot (\alpha - \delta)$  is equal to :
  - $q + r$
  - $q - r$
  - $-(q + r)$
  - $-(p + q + r)$
- The equation  $\frac{a(x-b)(x-c)}{(a-b)(a-c)} + \frac{b(x-c)(x-a)}{(b-c)(b-a)} + \frac{c(x-a)(x-b)}{(c-a)(c-b)} = x$  is satisfied by
  - No value of  $x$
  - Exactly two values of  $x$
  - Exactly three values of  $x$
  - All values of  $x$



**Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.**

## Answer Key

1. (1)
2. (3)
3. (2)
4. (2)
5. (2)

6. (1)
7. (2)
8. (3)
9. (3)
10. (4)



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